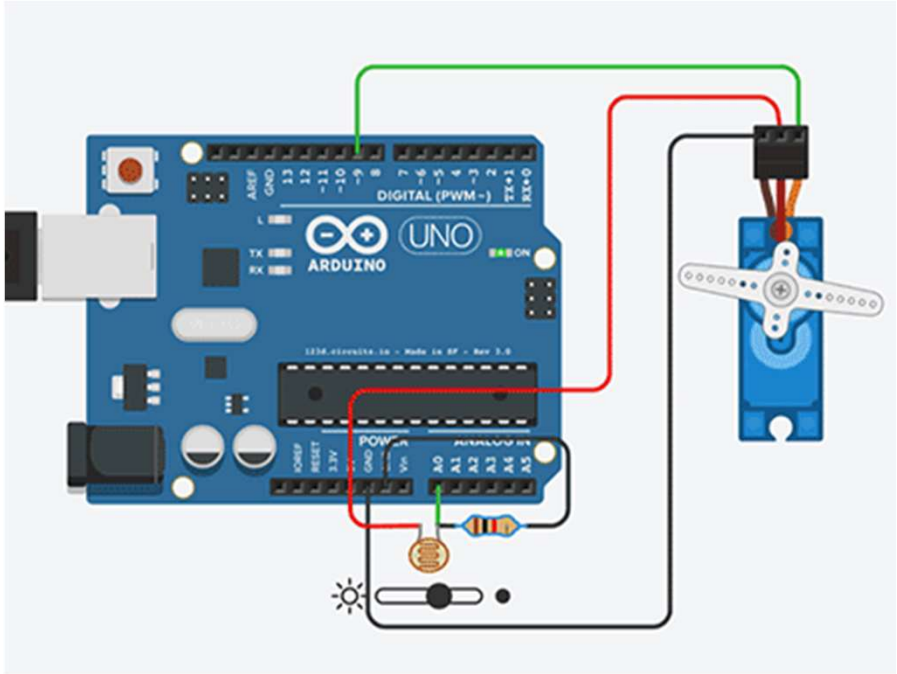
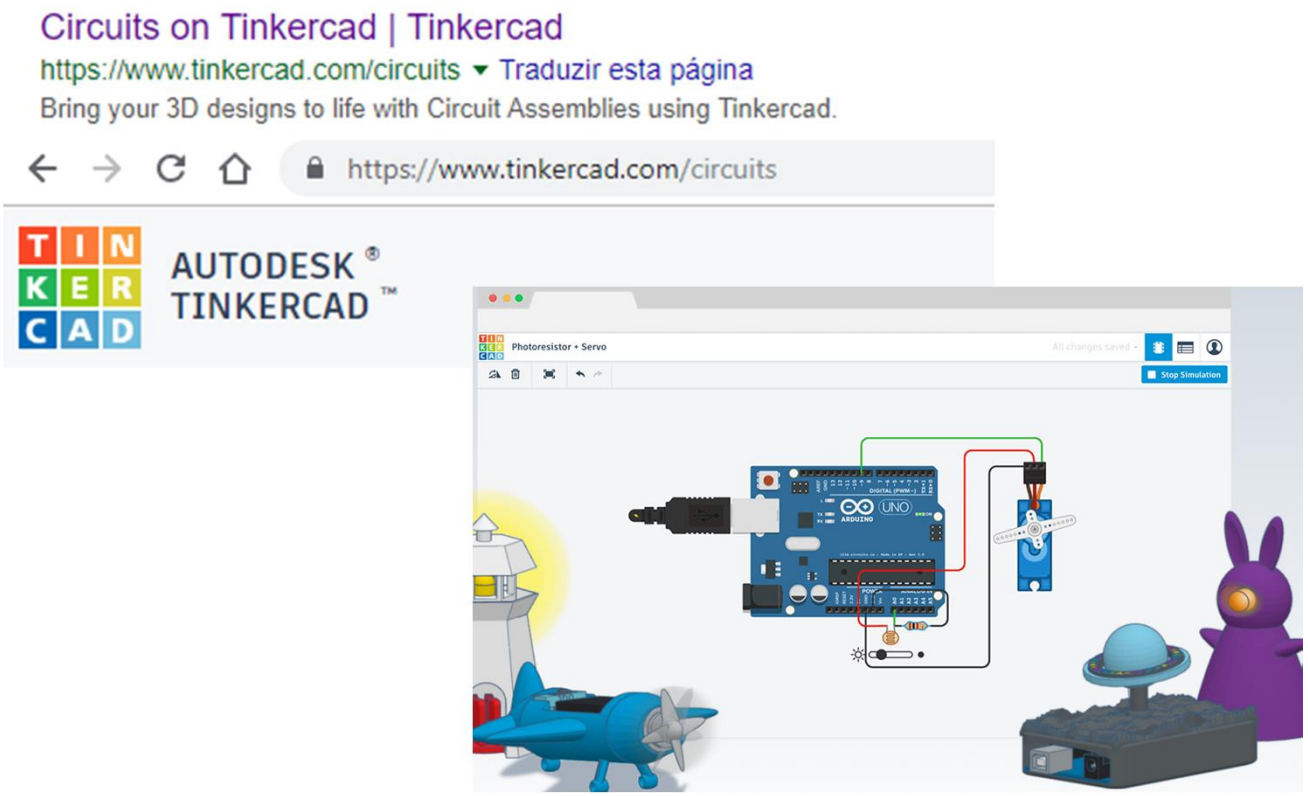
Simulação de Sistemas Embarcados



Objetivos:

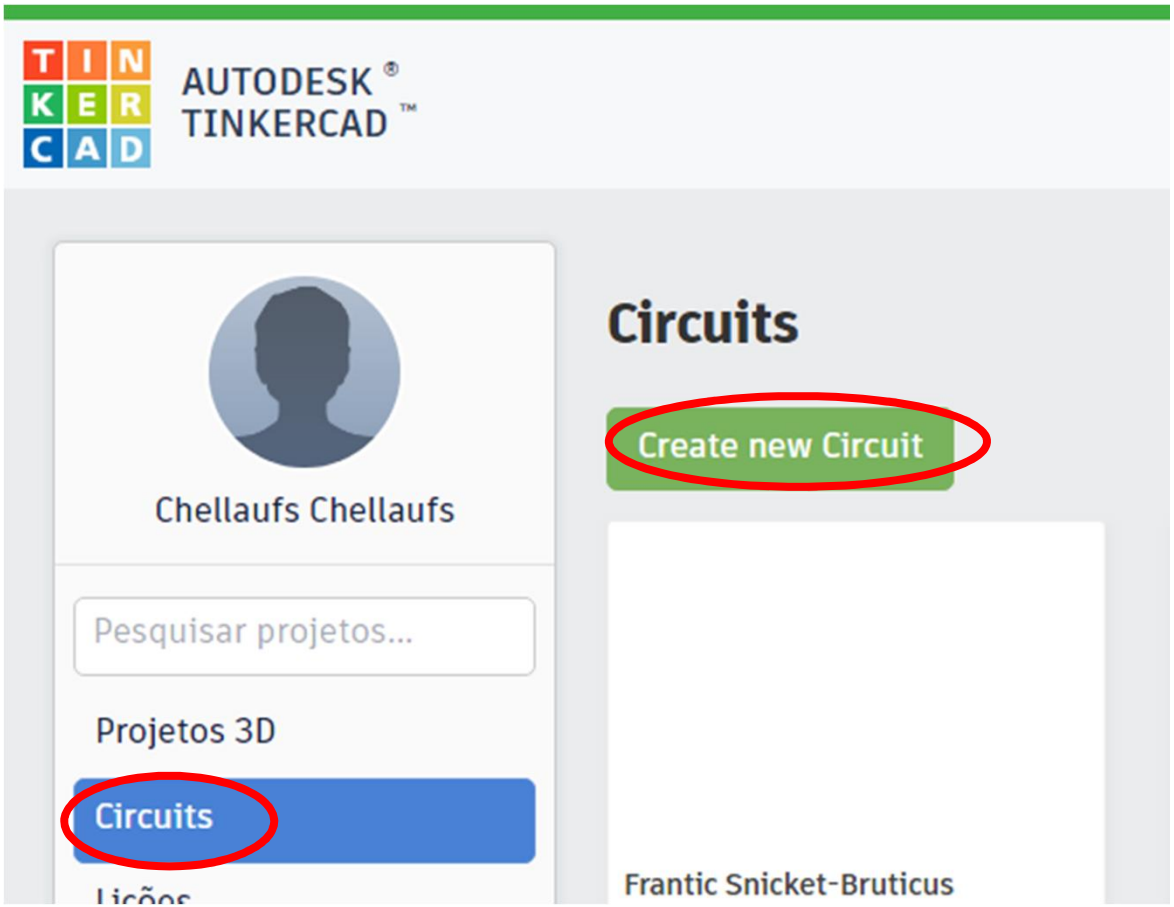
- Explorar recursos da ferramenta de simulação TINKERCAD
- Criar um projeto com microcontrolador
- Associar o código de firmware
- Executar e efetuar testes funcionais
- Compreender o uso de registradores e datasheet para adicionar novas funcionalidades ao firmware

Criar Conta ou efetuar o Login

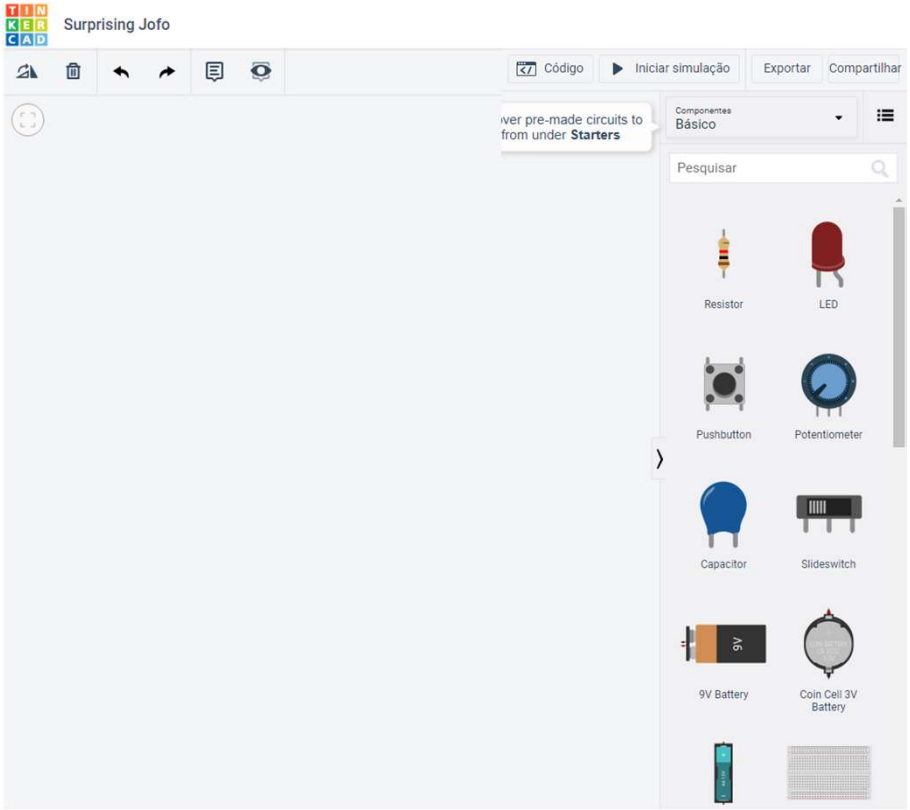


Criar Conta ou efetuar o Login

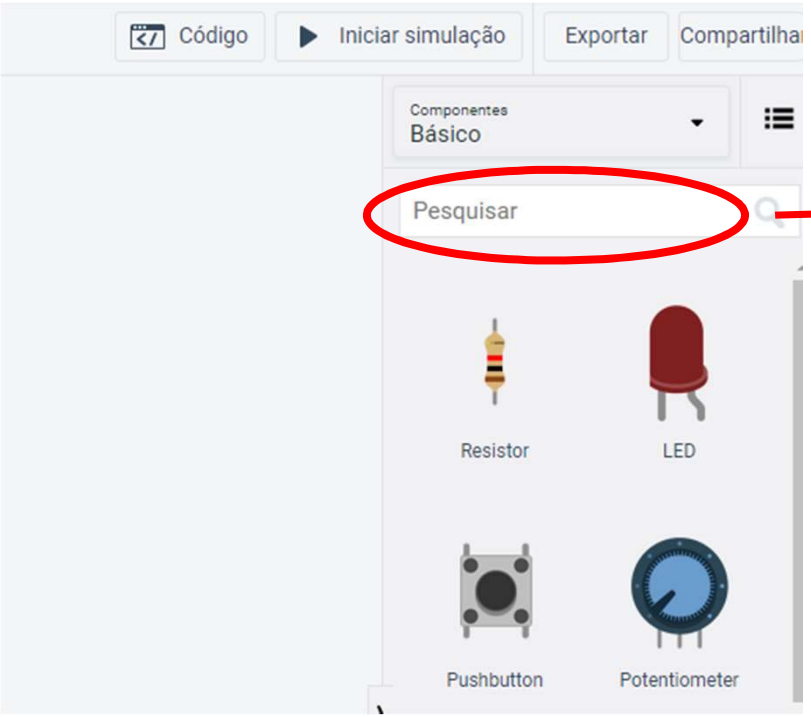
Criar Circuito

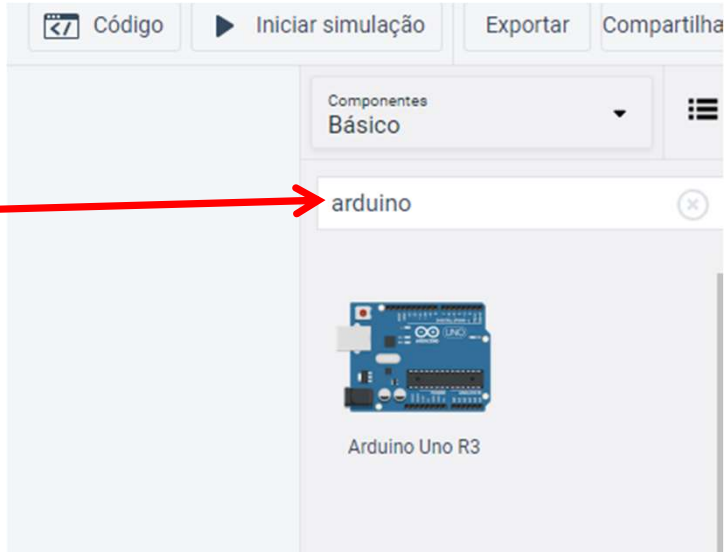


Criar Circuito : Tela inicial



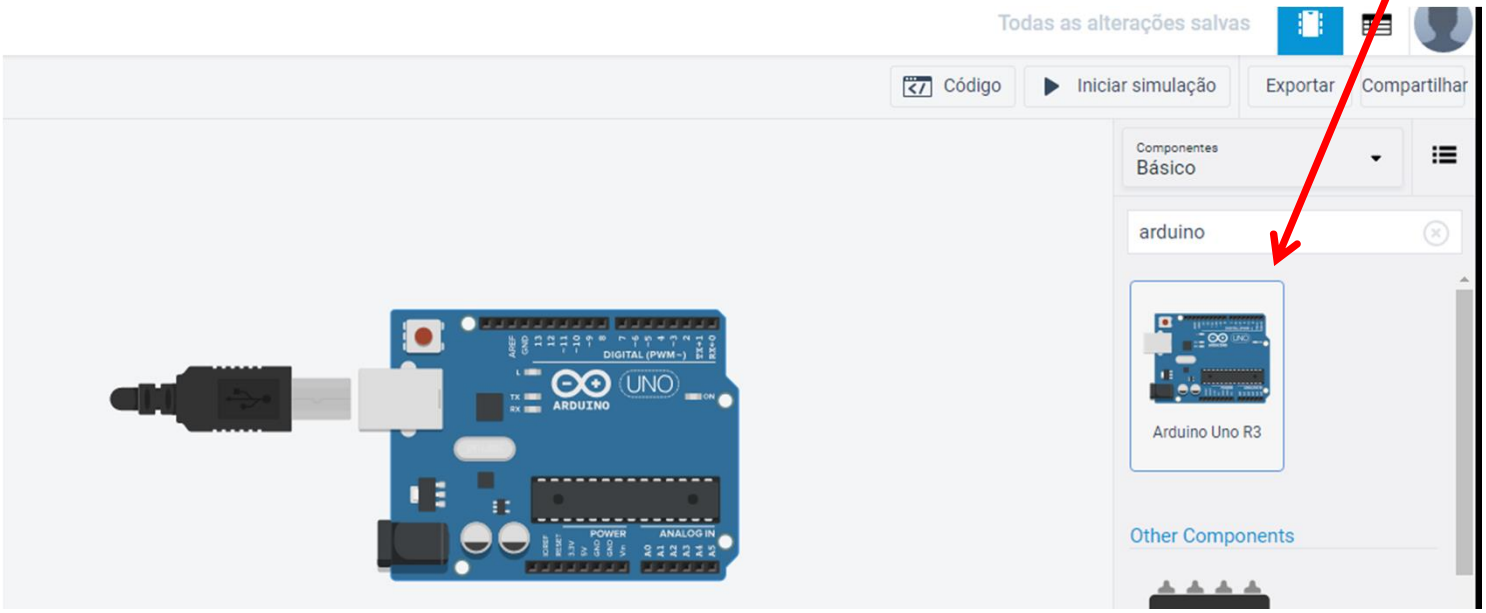
Criar Circuito





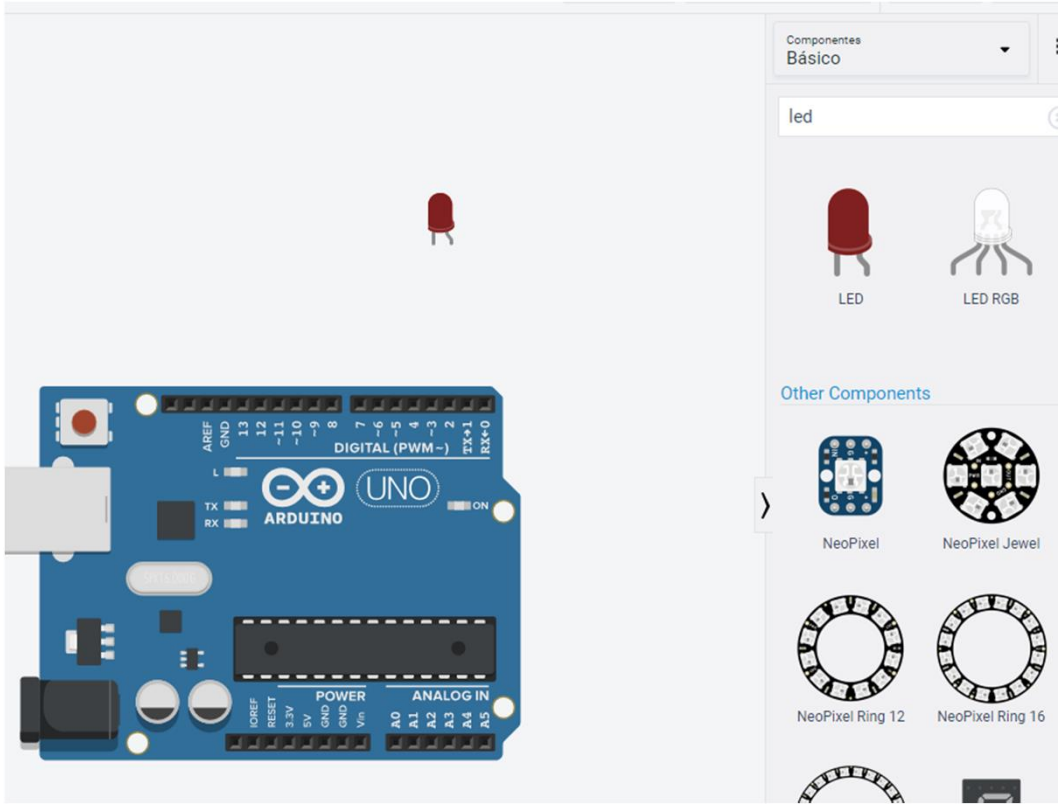
Criar Circuito

Clicar para selecionar



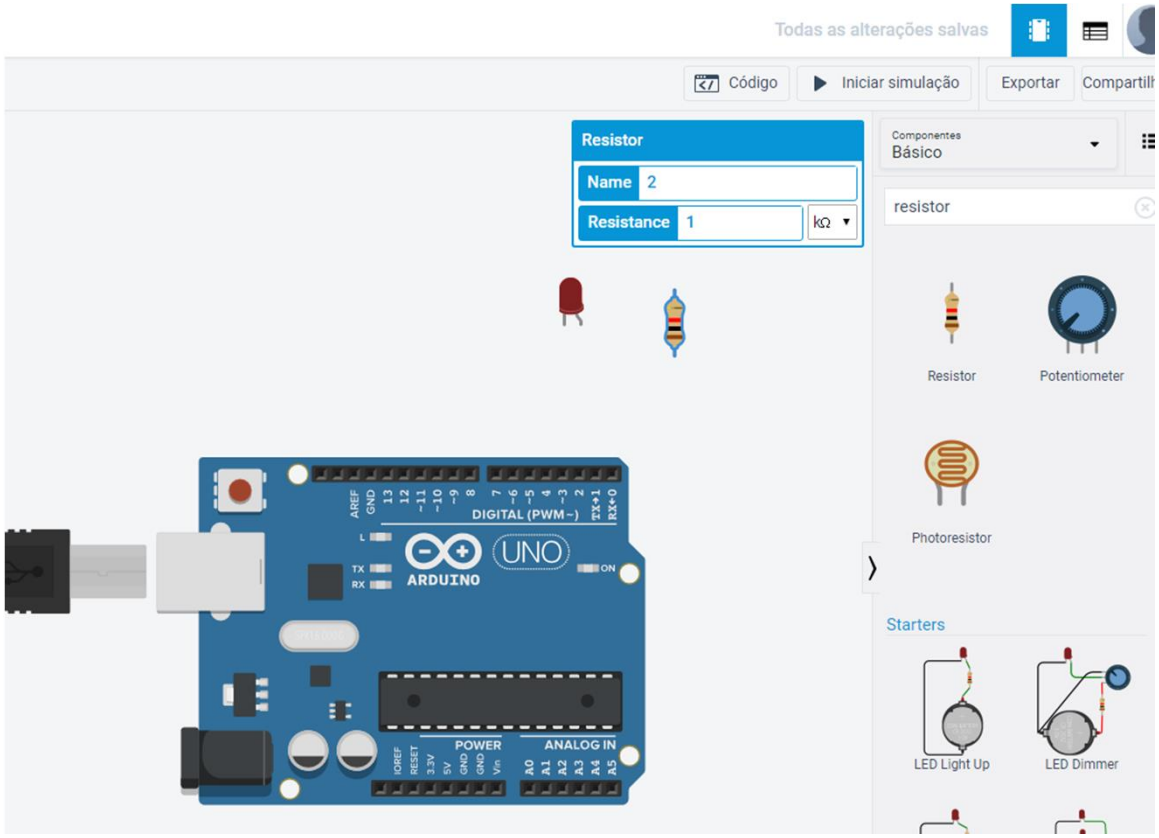
Criar Circuito

Adicionar um LED



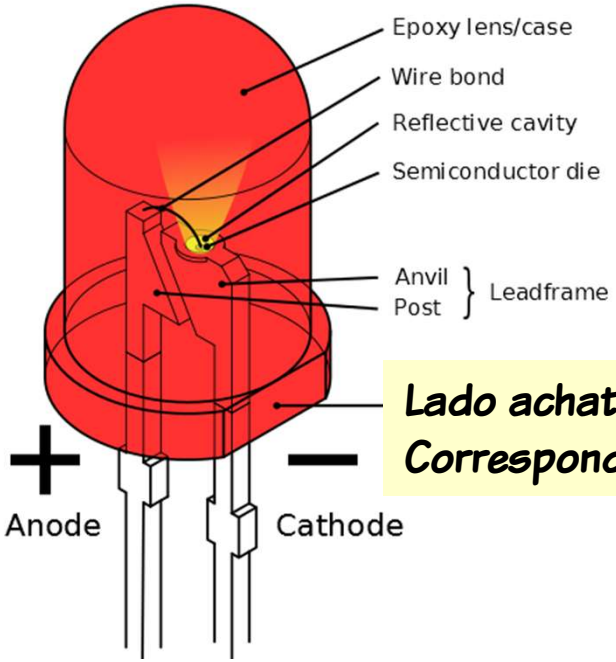
Criar Circuito

Adicionar um Resistor



Criar Circuito

LED é Polarizado

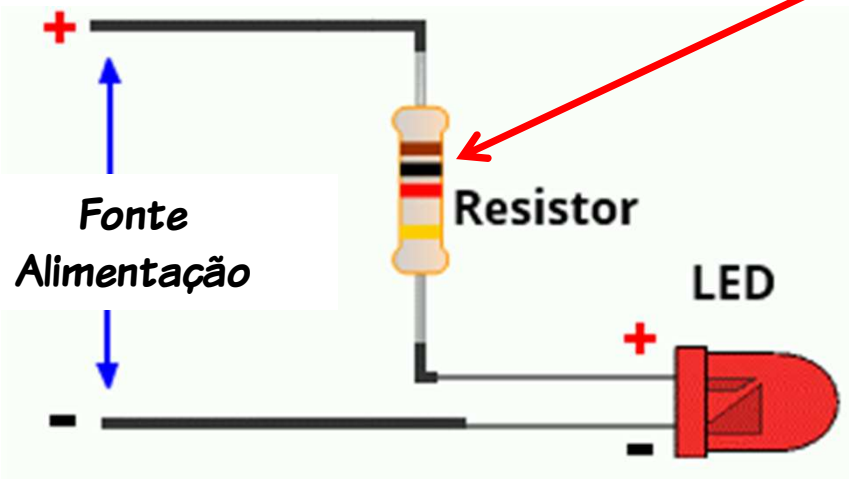


Lado achatado
Corresponde ao negativo da fonte (catodo)

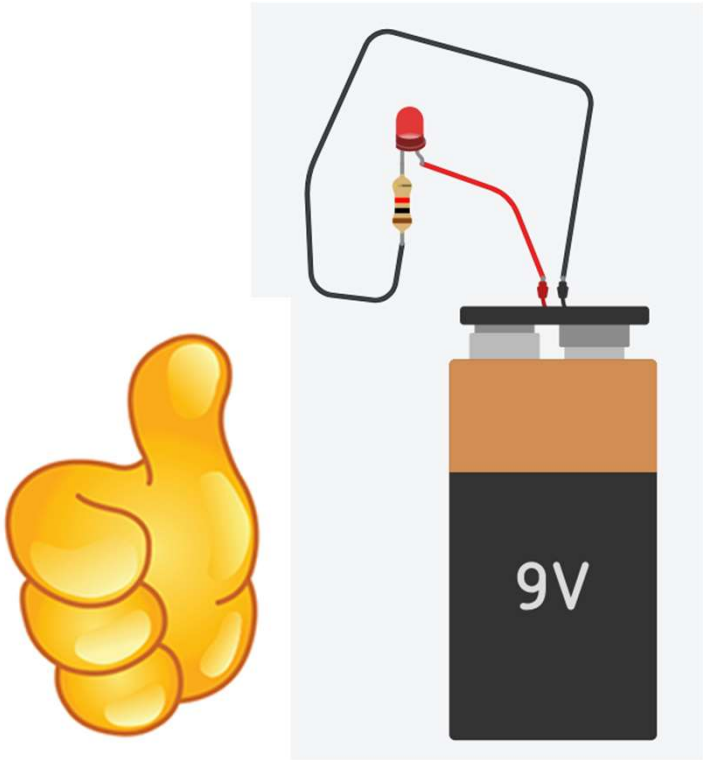
Criar Circuito

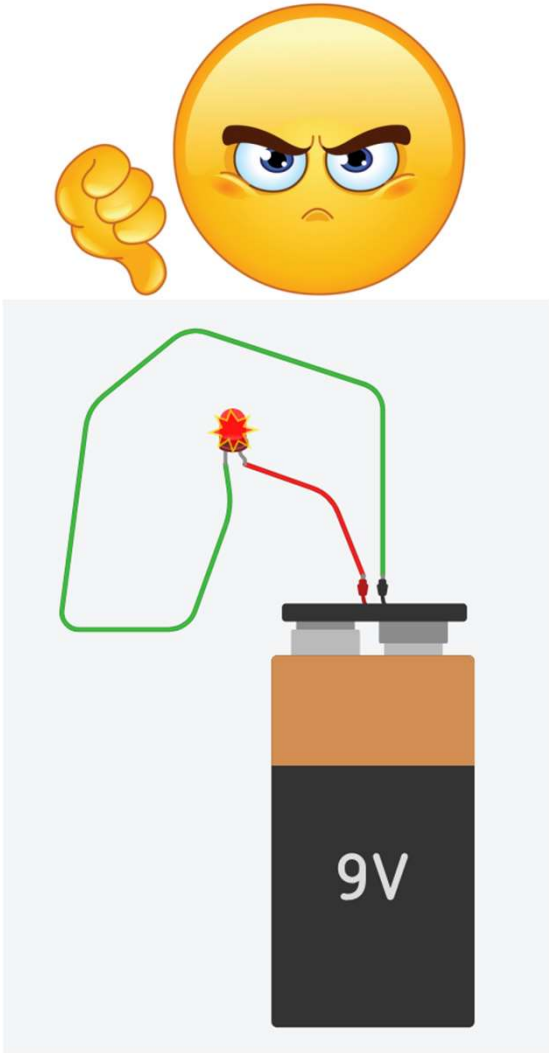
LED é Polarizado

Necessário
Resistor Limitador de Corrente

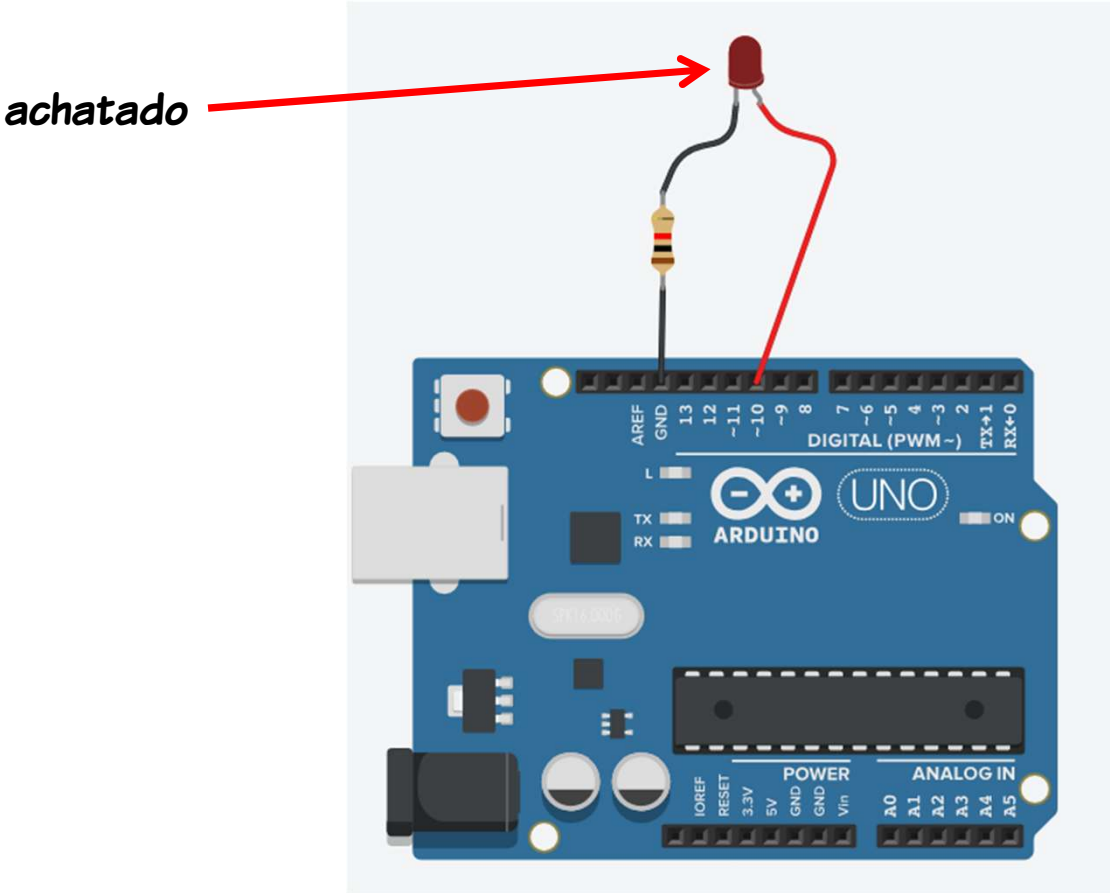


Criar Circuito





Criar Circuito: Conexão LED ao Arduino



Criar Circuito: Firmware

Criar Circuito: Firmware

Criar Circuito: Firmware

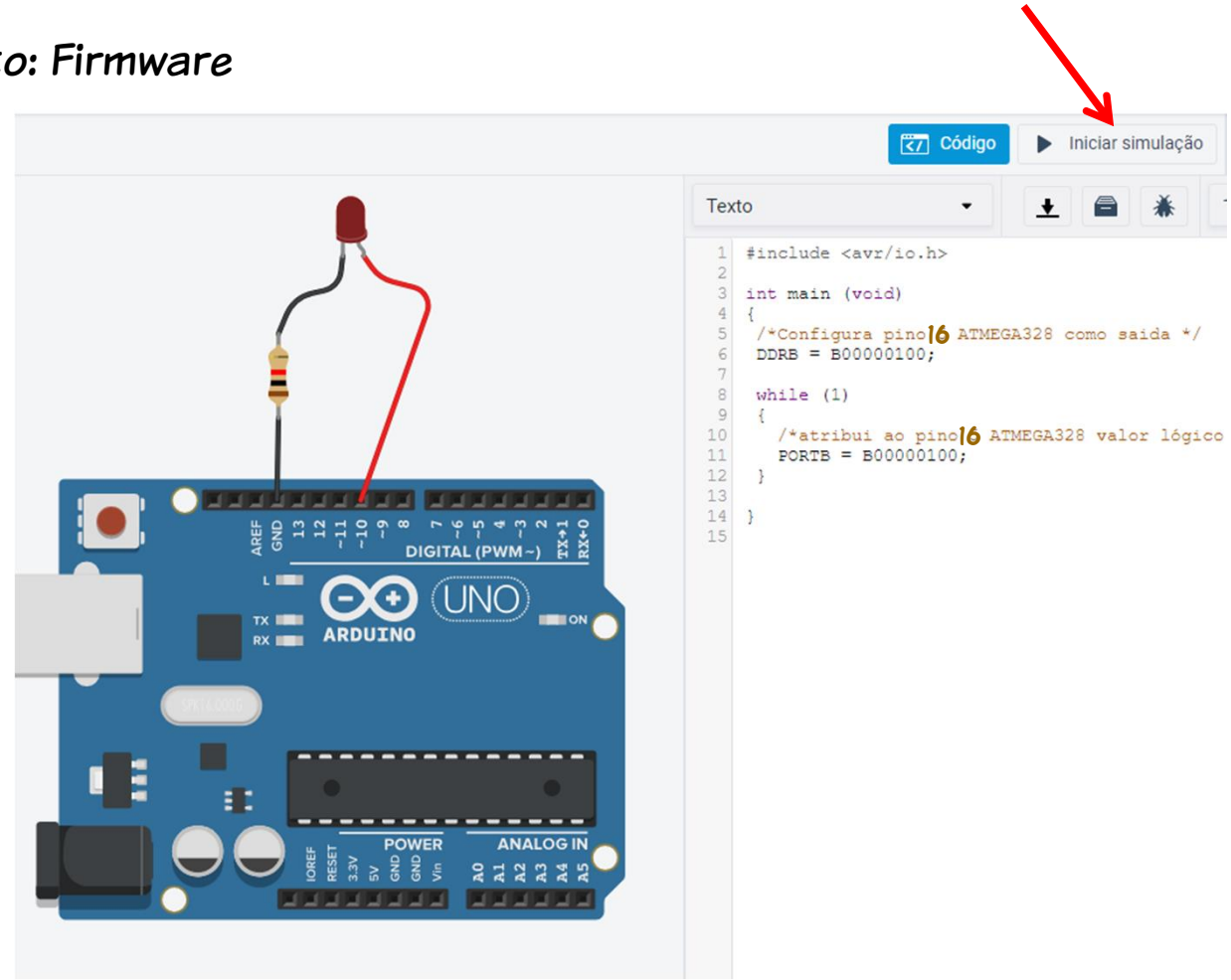
Todas as alterações salvas

Código | Iniciar simulação | Export

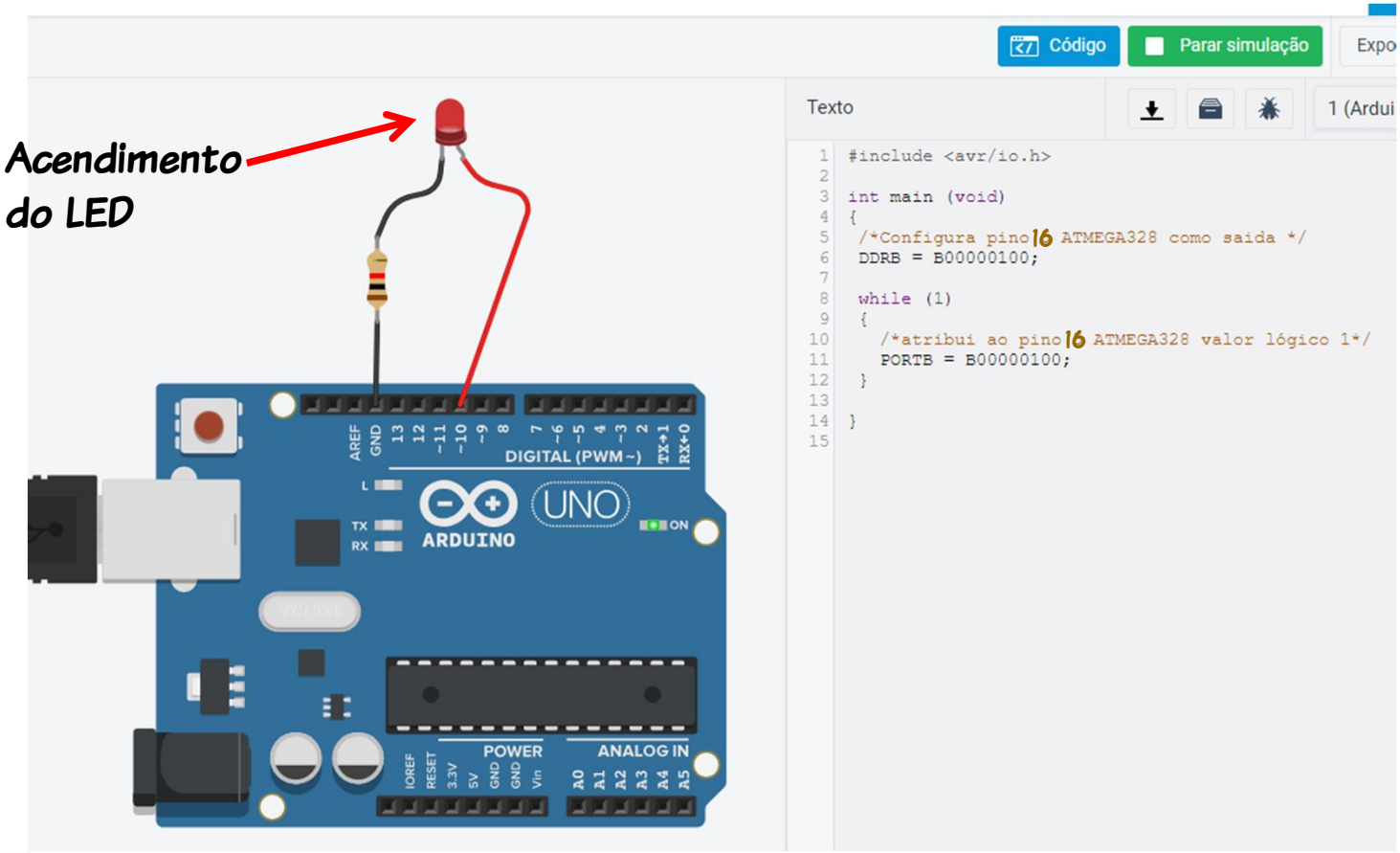
Texto | 1 (Arduin

```
#include <avr/io.h>

int main (void)
{
 /*Configura pino 16 ATMEGA328 como saida */
 DDRB = B00000100;

 while (1)
 {
   /*atribui ao pino 16 ATMEGA328 valor lógico 1*/
   PORTB = B00000100;
 }

}
```

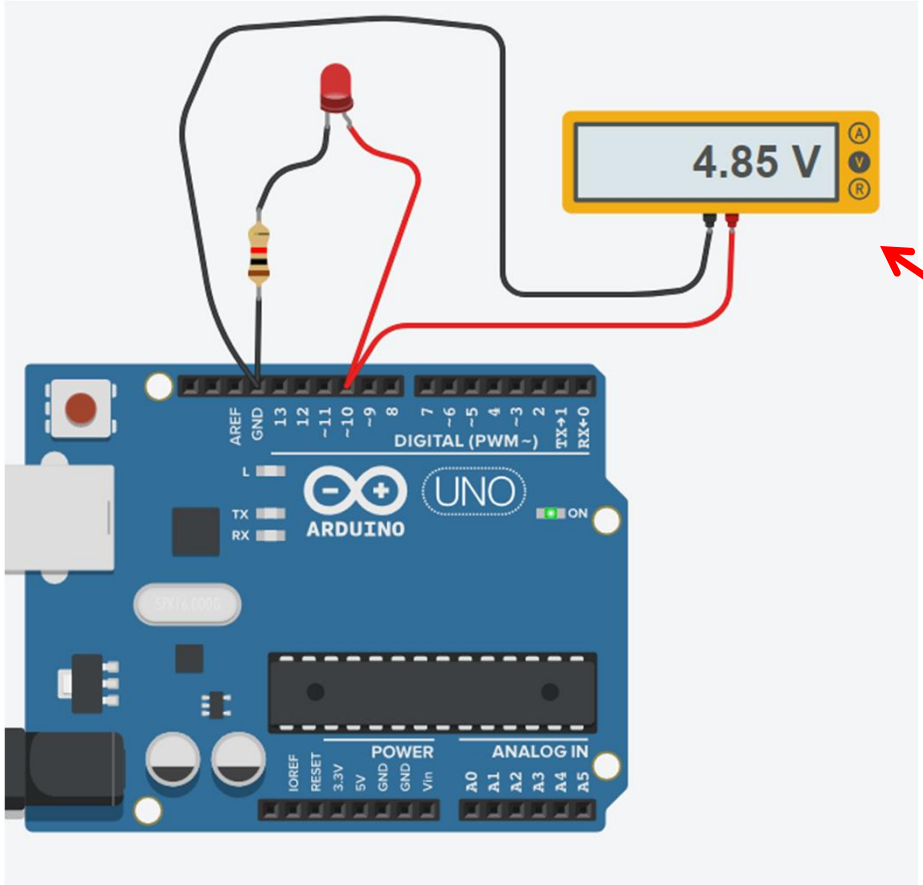
Criar Circuito: Firmware



Criar Circuito: Firmware



Criar Circuito: Firmware



Voltimetro
indica tensão para
nível lógico 1

Criar Circuito: Firmware - Funcionamento

Datasheet ATMEGA328

Atmel

ATmega328P

8-bit AVR Microcontroller with 32K Bytes In-System Programmable Flash

DATASHEET

Criar Circuito: Firmware - Funcionamento

Datasheet ATMEGA328

13.4.3 DDRB – The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

pg 72

```
int main (void)
{
 /*Configura pino 16 ATMEGA328 como saida */
 DDRB = B00000100;

 while (1)
 {
   /*atribui ao pino 16 ATMEGA328 valor lógico 1*/
   PORTB = B00000100;
 }

}
```

Criar Circuito: Firmware - Funcionamento

Datasheet ATMEGA328

13.4.2 PORTB – The Port B Data Register

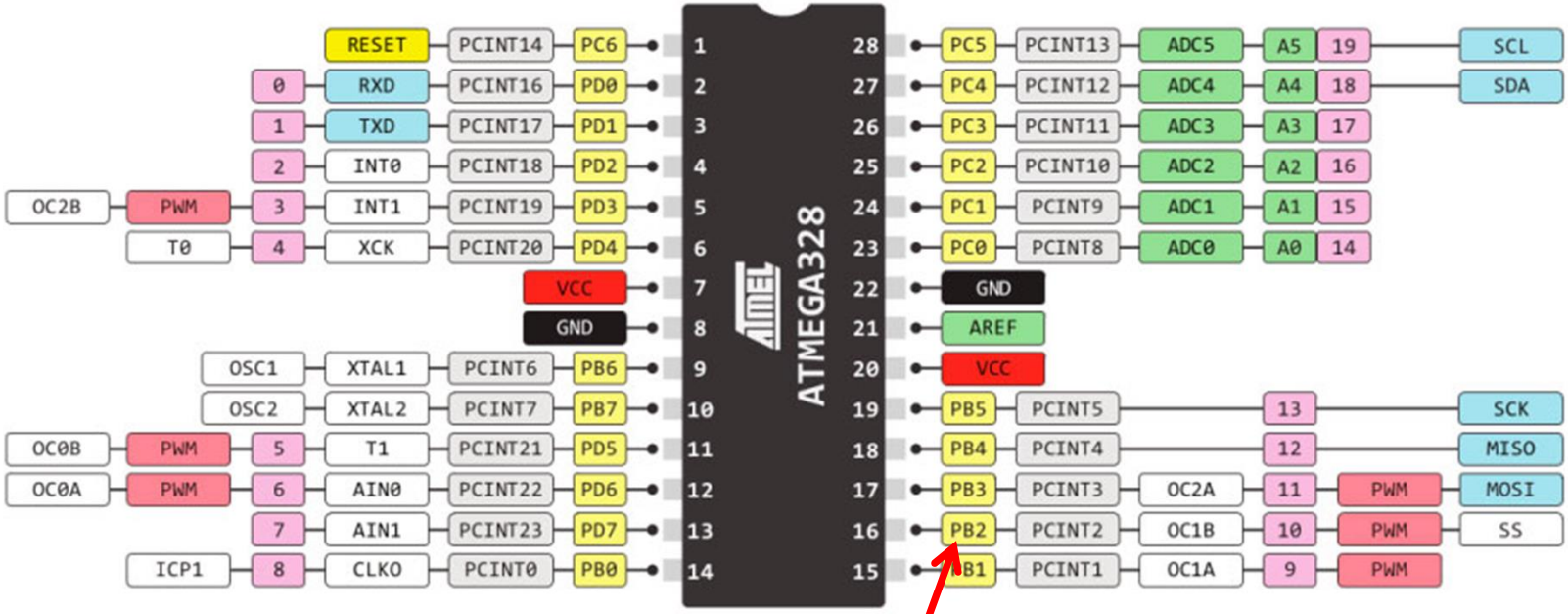
Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0						

pg 72

```
#include <avr/io.h>

int main (void)
{
 /*Configura pino 5 ATMEGA328 como saida */
 DDRB = B00000100;

 while (1)
 {
   /*atribui ao pino 5 ATMEGA328 valor lógico 1*/
   PORTB = B00000100;
 }

}
```

Criar Circuito: Firmware - Funcionamento

Datasheet ATMEGA328

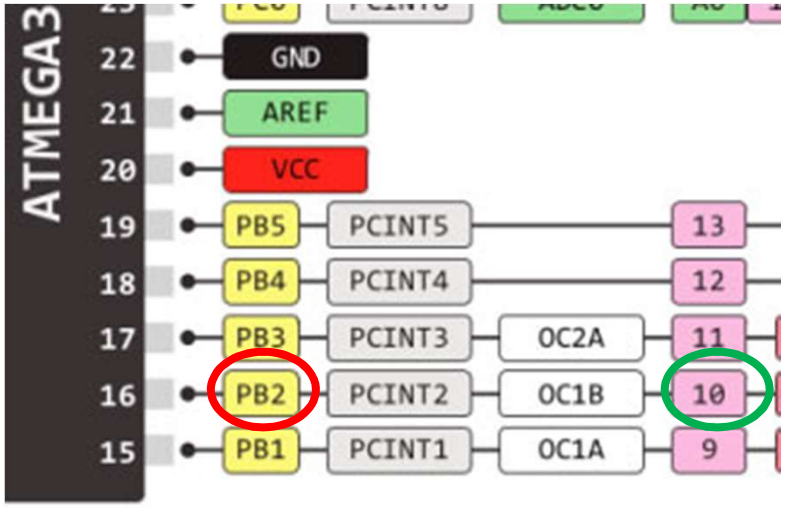


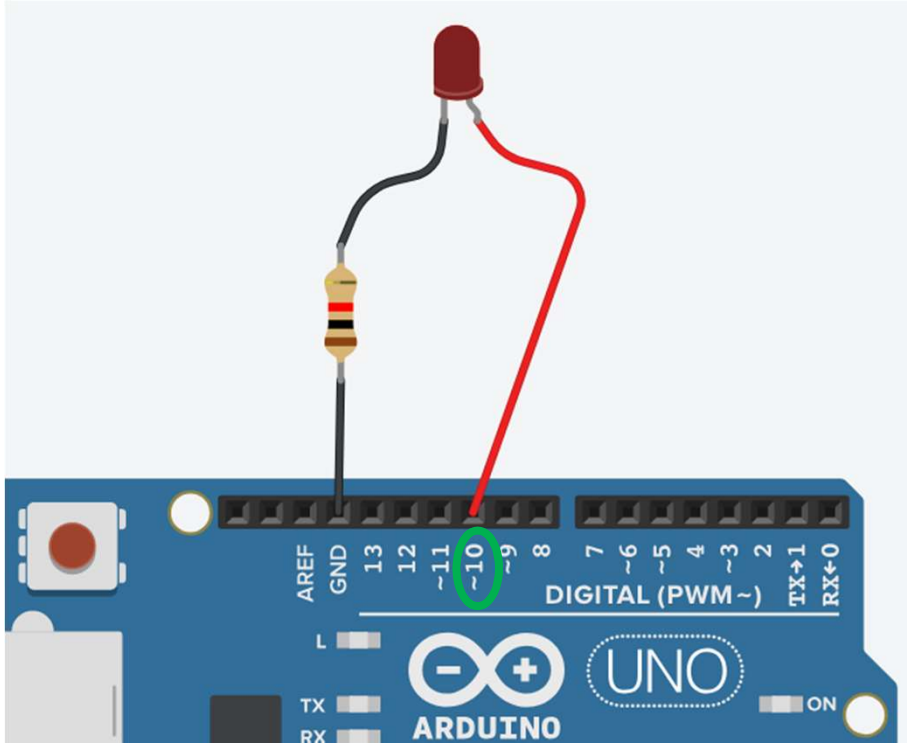
13.4.2 PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Criar Circuito: Firmware - Funcionamento

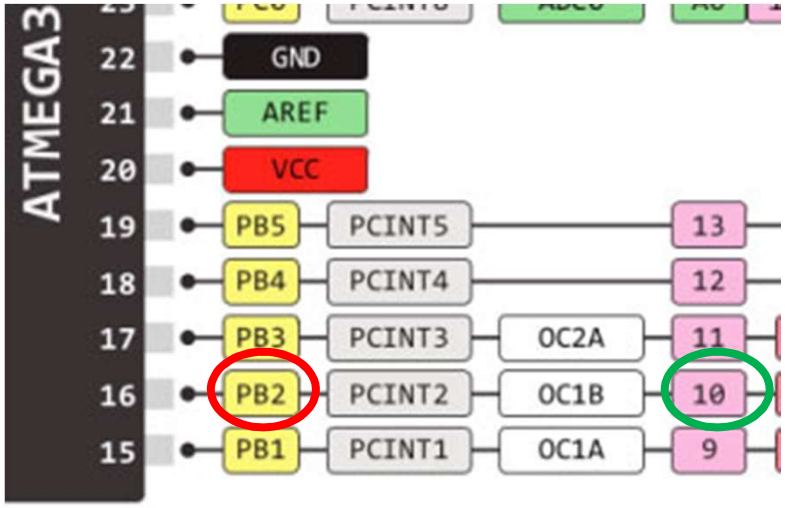
Datasheet ATMEGA328

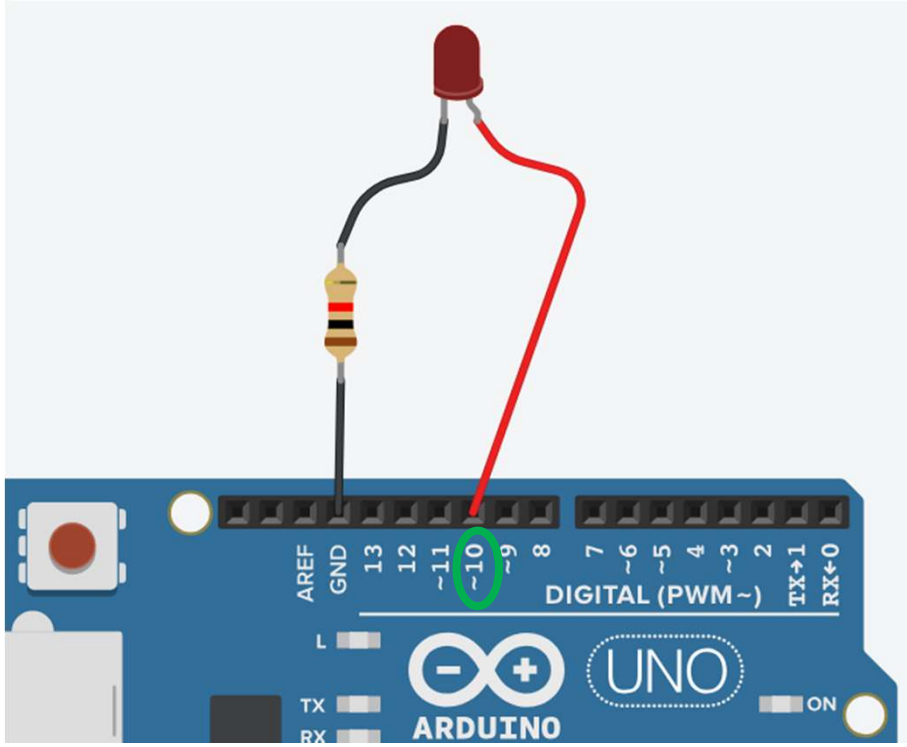




Criar Circuito: Firmware - Funcionamento

Datasheet ATMEGA328





Criar Circuito: Firmware - Funcionamento

Desafios

- Implementar firmware que configure e acione outros pinos
- Implementar firmware que acione os níveis 1 e 0 alternadamente (efeito de piscar). Dica: utilize função para delay da biblioteca **delay.h**
- Pesquise sobre a função millis() e faça uma implementação da alternância de acionamento do LED sem usar o delay

```
#include <util/delay.h>
```

```
_delay_ms(500);
```

Criar Circuito: Firmware – Funcionamento

Referências

ATmega328P
http://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P_Datasheet.pdf

Tinkercad
https://www.tinkercad.com/circuits